This document contains lecture notes from 18.212 (Algebraic Combinatorics), as taught in Spring 2025 by Professor Postnikov.

Lecture 1 (February 3)

The website for this class is https://math.mit.edu/~apost/courses/18.212/. There is no final exam; all grades will be based on problem sets, and participation is encouraged.

Catalan numbers

The **Catalan numbers** $\{C_n\}_{n\geq 0}$ are defined as such.

Definition.

C_n is the number of sequences $\{\varepsilon_1, \ldots, \varepsilon_{2n}\}$ such that:

- $\varepsilon_i = \pm 1$
- $\varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_i \geq 0 \,\forall\, i$
- $\varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_{2n} = 0.$

These are often represented as (A) **Dyck paths** (plotting the sums on the x-y plane. It goes up and down). (We are assigning letters to various things that C_n counts so that we can refer to them later.)

We can compute small values.

n	0	1	2	3	4	5	$\ldots$
C_n	1	1	2	5	14	42	$\ldots$

Theorem 1 (Catalan numbers).

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

Other applications

Catalan numbers count a lot of things!

Theorem 2 (Triangulations).

(B) C_n is the number of triangulations of a convex $n+2$-gon.

Theorem 3 (Parenthesizations).

(C) C_n is the number of parenthesizations of $n+1$ letters.

A **plane tree** is a rooted tree as drawn on the plane, so that the children of any vertex are ordered.

Note that in a **plane binary tree**, we have a special case that if a vertex only has one child, we record whether it is left or right.

Theorem 4 (Plane binary trees).

(D) C_n is the number of plane binary trees with n vertices.
(E) It is also the number of complete plane binary trees with $n+1$ leaves, i.e. those where every non-leaf vertex has exactly two children.

Theorem 5 (Plane trees).

(F) C_n is the number of plane trees with $n+1$ vertices.

Theorem 6 (Perfect matchings).

(G) C_n is the number of non-crossing perfect matchings on $2n$ vertices.
(H) The same is true for non-nesting perfect matchings.

Some bijection proofs

Theorem 7 (Triangulations (B) and plane binary trees (D)).

There is a natural bijection between triangulations of $n+2$-gons and plane binary trees with n vertices.

Proof. Number the vertices. Consider the triangle with the edge between 1 and $n+2$. This will be the root of our tree. Then it borders at most 2 triangles, which can be considered its children. We continue doing this and obtain a plane binary tree. The reverse is by construction.

Theorem 8 (Complete plane binary trees (E) and parenthesizations (C)).

There is a natural bijection between complete binary trees with $n+1$ leaves and parenthesizations of $n+1$ letters.

Proof. Each leaf is one letter. If two vertices share a parent, we draw a parenthesis around them (so that every non-leaf is a pair of parentheses). The reverse is by construction.

Theorem 9 (Plane trees (F) and Dyck paths (A)).

There is a natural bijection between plane trees with $n+1$ vertices and Dyck paths with $2n$ steps.

Proof. We use depth-first search traversal (go down the leftmost edge that hasn't been visited yet, going back up if you're at a leaf). Each edge is a step. Going down a new edge is an up step; going backwards up an edge is an up step.

One way to visualize the reverse is to "glue" each up step in a Dyck path to the next down step at the same height.

Lecture 2 (February 5)

We'll continue with more bijections and provide a proof of the formula $C_n = \frac{1}{n+1}\binom{2n}{n}$.

Theorem 10 (Plane binary trees (D) and complete plane binary trees (E)).

There is a natural bijection between plane binary trees with n vertices and complete plane binary trees with $n+1$ leaves.

Proof. If a vertex has only one child, add the other one. Also, add two children to each leaf of the original tree. (The inverse is by removing all of the leaves.)

We know it has $n+1$ leaves since we added enough vertices for every original vertex to have two children, so that there are $2n+1$ total.

Theorem 11 (Dyck paths (A) and non-crossing perfect matchings (G)).

There is a natural bijection between Dyck paths with $2n$ steps and non-crossing perfect matchings with $2n$ vertices.

Proof. Each step in the Dyck path is a perfect matching. Match each up step to the next down step at the same height. The inverse is not hard to see: go up if a vertex is before its partner; otherwise, go down.

Theorem 12 (Dyck paths (A) and non-nesting perfect matchings (H)).

There is a natural bijection between Dyck paths with $2n$ steps and non-nesting perfect matchings with $2n$ vertices.

Proof. The inverse is the same as above. The forward direction is a bit harder and was not covered in class.

We now have two connected components of objects counted by the Catalan numbers (where each edge is a bijection): {A, F, G, H} and {B, D, E, C}. It would be great if we could connect them.

Problem (Dyck paths (A) and parenthesizations (C)).

Find a bijection between Dyck paths and parenthesizations.

Note that the most obvious choice (each left parenthesis = up, right parenthesis = down) does not work.

Proof 1 of C_n (Reflection method)

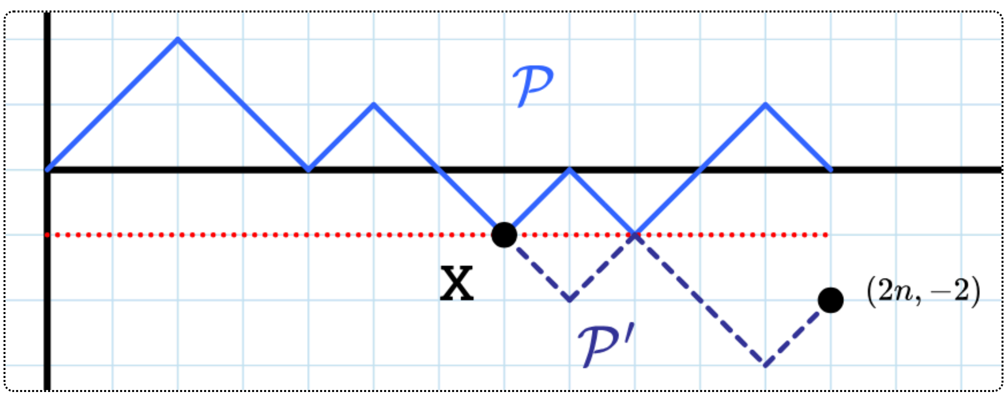
Let's prove Theorem 1.

Theorem 1 (Catalan numbers).

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

We will prove that this is the number of Dyck paths with $2n$ steps.

Proof 1 (Reflection method).



We call a "lattice path" from $(0,0)$ to $(2n,0)$ any path consisting of either up-right or down-right moves. Note that the total number of these is $\binom{2n}{n}$, since n must be up moves and n must be down moves.

Note that C_n is thus $\binom{2n}{n}$ minus the number of "bad paths", i.e. ones that dip below the x-axis. Those paths are exactly the ones that intersect $y=-1$. Suppose $X=(x,-1)$ is the leftmost intersection of our path $\mathcal{P}$ with $y=-1$.

Let $\mathcal{P}'$ be the path obtained by reflecting the portion of $\mathcal{P}$ after X around the line $y=-1$. The new endpoint of this path is $(2n,-2)$.

Claim (Enumerating bad lattice paths).

The map $\mathcal{P} \mapsto \mathcal{P}'$ is a bijection between bad lattice paths from $(0,0)$ to $(2n,0)$ and <u>all</u> lattice paths from $(0,0)$ to $(2n-2)$.

Proof of Claim. Find the first intersection with $y=-1$ and reflect.

Therefore, the number of Dyck paths is

$$\begin{aligned}\binom{2n}{n}-\binom{2n}{n-1} &= \frac{(2n)!}{n!n!}-\frac{2n!}{(n-1)!(n+1)!} \\ &= \frac{(2n)!}{n!(n+1)!}(n+1-n)=\frac{1}{n+1}\binom{2n}{n}.\end{aligned}$$ □

Lecture 3 (February 7)

Our previous proof required us to use complementary counting and do some subtraction/algebra. It would be nice if we didn't do that and could use a purely combinatorial argument. The easiest way to interpret the $\frac{1}{n+1}$ term is like this:

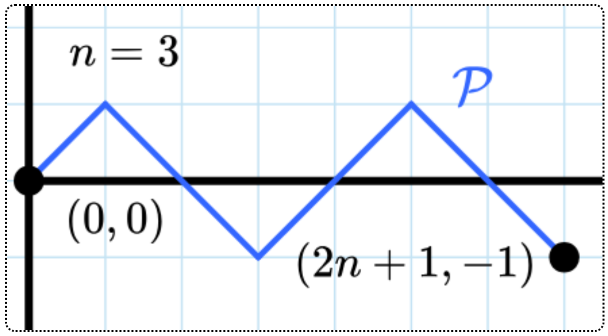
Problem (Can we find a "subtraction-free" proof?).

Can we subdivide all $\binom{2n}{n}$ lattice paths with n up and n down steps into a disjoint union of blocks of size $n+1$, each of which contains exactly one Dyck path?

We'll first use a slight modification to our formula for C_n.

$$\frac{1}{n+1}\binom{2n}{n}=\frac{(2n)!}{n!(n+1)!}=\frac{1}{2n+1}\binom{2n+1}{n}$$

$\binom{2n+1}{n}$ is now the number of all lattice paths with n up steps and $n+1$ down steps, so that they land at $(2n+1,-1)$.

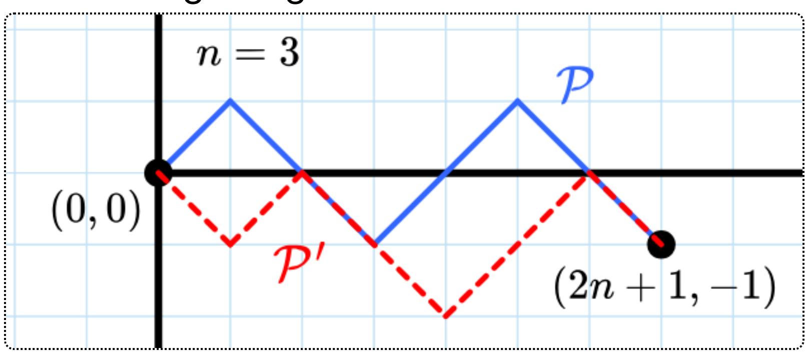


We define an **augmented Dyck path** as a Dyck path with an extra down step at the end. Evidently, there are exactly as many of these as there are Dyck paths.

Can we partition the set of lattice paths to $(2n+1, -1)$ into groups of $2n+1$ in which one is an augmented Dyck path?

Proof 2 of C_n (Cyclic shifts)

Given a lattice path $\mathcal{P}$, we define the operation $c : \mathcal{P} \mapsto \mathcal{P}'$ obtained by moving the last step of $\mathcal{P}$ to the beginning.



If we visualize a path by writing the steps as a sequence of ± 1s, such as $\mathcal{P} = (1, -1, -1, 1, 1, -1, -1)$, it's clear what we're doing: move the -1 from the back of $\mathcal{P}$ onto the front to obtain $\mathcal{P}' = (-1, 1, -1, -1, 1, 1, -1)$.

This is called a "cyclic shift" because it's an action of the cyclic group of order $2n+1$, $\mathbb{Z}/(2n+1)\mathbb{Z}$, onto the lattice paths. (Don't worry if you don't know what this means). You can also imagine writing the ± 1s of the sequence onto a circle and rotating it.

If we repeatedly apply c onto a path $\mathcal{P}$, we can obtain at most $2n+1$ different paths (before we return to the original $\mathcal{P}$). We call this the "orbit" of $\mathcal{P}$ (from group theory) or a "block".

The orbit of our above example $\mathcal{P}$ is

$$\begin{aligned}
\{(&\ \ 1,-1,-1,\ \ \ 1,\ \ \ 1,-1,-1),\\
(&-1,\ \ \ 1,-1,-1,\ \ \ 1,\ \ \ 1,-1),\\
(&-1,-1,\ \ \ 1,-1,-1,\ \ \ 1,\ \ \ 1),\\
(&\ \ 1,-1,-1,\ \ \ 1,-1,-1,\ \ \ 1),\\
(&\ \ 1,\ \ \ 1,-1,-1,\ \ \ 1,-1,-1),\\
(&-1,\ \ \ 1,\ \ \ 1,-1,-1,\ \ \ 1,-1),\\
(&-1,-1,\ \ \ 1,\ \ \ 1,-1,-1,\ \ \ 1)\}
\end{aligned}$$

In fact, we make the following two claims (that would prove the theorem):

Claim 1 (Order of orbits).

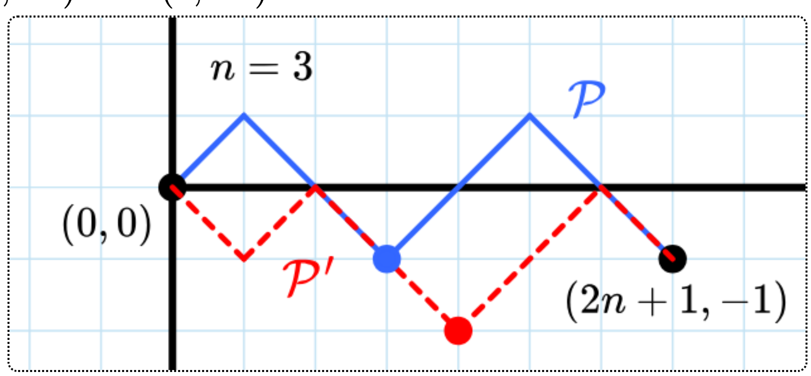
Each orbit contains exactly $2n+1$ paths. (In other words, all $2n+1$ paths $\mathcal{P}$, $\mathcal{P}'$, $\mathcal{P}''$, etc. are distinct.)

Claim 2 (Augmented Dyck paths in orbits).

Each orbit contains exactly one augmented Dyck path.

We can prove these claims by considering $m(\mathcal{P})$, the position of the first minimum of $\mathcal{P}$.

For example, in the below picture, $m(\mathcal{P}) = 3$ and $m(\mathcal{P}') = 4$, since their first minima are at $(3, -1)$ and $(4, -2)$.



We can make two quick observations:

1. $m(\mathcal{P}) \in \{1, 2, 3, \ldots, 2n+1\}$. Clearly position 0 cannot be a minimum since $(0, 0)$ is at the very least higher than $(2n+1, -1)$.

2. $\mathcal{P}$ is an augmented Dyck path if and only if $m(\mathcal{P}) = 2n+1$. This is because both mean that the path never reaches $y = -1$ before $x = 2n+1$.

Now, we would like to prove the following, which would imply both claims using our observations:

Claim 3 ($m(\mathcal{P})$ is cyclical).

As we apply cyclic shifts $\mathcal{P} \xrightarrow{c} \mathcal{P}' \xrightarrow{c} \mathcal{P}'' \xrightarrow{c} \cdots \xrightarrow{c} \mathcal{P}$, $m(\mathcal{P})$ changes cyclically $i \to i+1 \to i+2 \to \cdots \to 2n+1 \to 1 \to 2 \to \cdots \to i-1$.

This is not really surprising since applying c "shifts" everything, including the minimum, one point to the right (we just have to prove that we don't accidentally create another minimum to the left of it). The wraparound is since we go from an augmented Dyck path (where $m(\mathcal{P}) = 2n+1$) to a down step followed by a Dyck path (which would indeed have $m(\mathcal{P}) = 1$.

The full proof is left as an exercise. There are some subtle corner cases where you have to use the fact that n and $2n+1$ are coprime.

Sidenote

There is a standard proof of the Catalan numbers using generating functions, but it will not be covered in the class this year. It can be found online, e.g. in the past lecture notes.

Sorting of permutations

Let $w = (w_1, w_2, \ldots, w_n)$ be a **permutation** of $\{1, 2, \ldots, n\}$ (i.e. for each $i \in [1, n]$, exactly one w_j equals i).

The **symmetric group** S_n is the set of all permutations of $\{1, 2, \ldots, n\}$.

Stack sorting

A **stack** (from CS) is a last-in-first-out (LIFO) data structure. It has two operations:

- "push", adding an item to the top of the stack, and
- "pop", removing and retrieving the topmost item of the stack.
  (Imagine a physical stack of books or coffee cups.)

We want to sort a permutation using a series of stack operations.

Example (Stack-sortability).

Suppose we have the permutation $4, 2, 1, 3$. We can sort this stack by doing the following:

- push 4 (input: $2\ 1\ 3$, stack: 4)
- push 2 (input: $1\ 3$, stack: $2\ 4$)
- push 1 (input: 3, stack: $1\ 2\ 4$)
- pop (input: 3, stack: $2\ 4$, output: 1)
- pop (input: 3, stack: 4, output: $1\ 2$)
- push 3 (stack: $3\ 4$, output: $1\ 2$)
- pop twice (output: $1\ 2\ 3\ 4$)

So, this permutation is **stack-sortable**.

Are all permutations stack-sortable? No: consider $2, 3, 1$; we're forced to push them all, and after popping 1 we are forced to pop 3.

How many stack-sortable permutations are there? As you may have guessed, we have:

Theorem 13 (Number of stack-sortable permutations).

The number of stack-sortable permutations in S_n is C_n.

The following theorem will help.

Theorem 14 ($2\ 3\ 1$-avoiding permutations).

Stack-sortable permutations are exactly those that are "$2\ 3\ 1$-avoiding", i.e. $\nexists\, i < j < k$ such that $w_k < w_i < w_j$ (the same relative order as $2\ 3\ 1$.)

Queue-sortable permutations

In the last few minutes, we'll mention queue-sortable permutations. A queue is first-in-first-out, so you can:

- push an element into the left
- pop an element from the right.

This isn't very interesting for sorting: a permutation will always come out the same way as it came in. However, let's allow a third operation: letting a specific number skip the queue and immediately go into the end. (Imagine a driver in traffic deciding to be a bad guy and jumping in front of everyone by using the lane in the opposite direction.)

Example (Queue-sortability).

The permutation $1, 3, 4, 2, 6, 5$ is **queue-sortable**:

- push three times (input $2\ 6\ 5$, queue $4\ 3\ 1$)
- pop (input $2\ 6\ 5$, queue $4\ 3$, output 1)
- let 2 jump the queue (input $6\ 5$, queue $4\ 3$, output $1\ 2$)
- pop twice (input $6\ 5$, output $1\ 2\ 3\ 4$)
- push (input 5, queue 6, output $1\ 2\ 3\ 4$)
- let 5 jump the queue (queue 6, output $1\ 2\ 3\ 4\ 5$)
- pop (output $1\ 2\ 3\ 4\ 5\ 6$)

We have some analogous theorems.

Theorem 15 (Number of queue-sortable permutations).

The number of queue-sortable permutations in S_n is C_n (allowing queue jumps).

Theorem 16 ($3\ 2\ 1$-avoiding permutations).

Queue-sortable permutations are exactly those that are "$3\ 2\ 1$-avoiding", i.e. $\nexists\, i < j < k$ such that $w_k < w_j < w_i$ (the same relative order as $3\ 2\ 1$.)